SPIRIT IN PHYSICS: Spirit as a Thermodynamic Information Quantity

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Abstract

Philosophical accounts often invoke "spirit" as a non-material essence, yet have lacked an operational definition compatible with physics. Here we introduce Spirit as a thermodynamic information quantity defined on a high-dimensional manifold that couples linguistic, behavioural and physiological signals. This paper presents a theoretical framework for Spirit, a model for interpreting its internal structures, and an experimental methodology for its measurement.

1. Introduction

1.1. Why "Spirit": The Negation of the Individual

The deliberate choice of "Spirit" over "Mind" in this paper stems from our fundamental ontological claim.

Traditional psychology and philosophy have conceptualized the human "mind" as an **internal entity, separate and independent** from the body, environment, and others. This Cartesian dualism—mind-body dualism—presupposes the concept of the "individual" (literally, the "undividable") and treats human existence as a closed system.

However, the framework presented in this paper fundamentally rejects this premise. We assert the following:

1. **Non-separability**: The human "mind" is not separate or independent from the field, physical environment, body, or others.
2. **Existence as an Open System**: Human beings are interactively **open systems**, constantly exchanging information and energy with their environment.
3. **Thermodynamic Information Field**: Human existence should be understood not as a binary opposition of "mind and body" but as an integrated **thermodynamic information field of field, physics, and information**.

From this perspective, the term "Spirit" was chosen to denote not a "mind" confined within an individual, but an **open informational structure that interpenetrates the environment, others, and physical fields**. As suggested by the Latin etymology of "Spirit"—*spiritus* (breath, wind)—Spirit is not a fixed entity but a **dynamic process**, fluid and unified with the field.

1.2. The Conceptual Gap and Its Resolution

The concept of "spirit" has long been central to philosophical and psychological inquiry, yet it has remained largely outside the scope of empirical science due to the absence of a physically grounded, measurable definition. This conceptual gap has hindered our ability to investigate the material basis of what are often considered non-material phenomena. To bridge this gap, we propose a fundamental re-conceptualization of Spirit, moving it from the metaphysical realm to the physical.

Our approach is built on a foundational premise established in modern physics: information is physical (Landauer, 1991). The erasure of one bit of information corresponds to a minimal dissipation of energy, a principle that has been experimentally verified (Bérut et al., 2012). If Spirit is fundamentally informational in nature, as we postulate, then it must also be subject to physical laws.

In this paper, we define Spirit as a thermodynamic information quantity. Specifically, we model it as a state on a high-dimensional manifold—termed Complex Space—that integrates multiple streams of data: linguistic (word associations), behavioural (reaction times), and physiological (skin potential, emotional expression). We then propose an experimental method to quantify this state and analyze its structure. By operationalizing Spirit in this way, we provide a new framework for investigating the physical underpinnings of consciousness and psychosomatic phenomena.

2. Theoretical Framework

2.1. The Physical Nature of Information

Our framework rests on the principle that information is a physical quantity, inextricably linked to thermodynamics. Landauer's principle states that any logically irreversible manipulation of information, such as the erasure of a bit, must be accompanied by a corresponding entropy increase in the non-information-bearing degrees of freedom of the information-processing apparatus. This establishes a direct connection between information theory and thermodynamics, which we extend to the concept of Spirit.

2.2. The Physical Definition of Spirit

We define Spirit not as a monolithic entity but as a dynamic state within a high-dimensional vector space. This "Spirit Physical Space" is a manifold constructed from informational and biological components.

Let the state of the Spirit, S, be a point in a space composed of a set of informational vertices V, edges E, and time axis T.

$$S = \{V, E, T\}$$

The energy potential of an edge connecting two informational vertices (w_I, w_O) is defined as the negative logarithm of their association probability, reflecting the information content or "surprise" of their connection:

$$E = -\ln P(w_O|w_I),\ V = \{\vec{w}_I, \vec{w}_O, ...\},\ T = \text{time axis}$$

We can then define Spirit, $\psi(S)$, as a physical field quantity—the functional derivative of the total information energy of the system with respect to its state:

$$\psi(S) = \frac{\delta E(S)}{\delta S}$$

2.3. The Open System Dynamics of Spirit

The Spirit operates as a thermodynamic open system, constantly exchanging information and energy with its environment. This mathematically expresses the central claim of this paper: that the "individual" is not a closed system but an open system in constant interaction with the external world.

The total entropy change of the system can be expressed as:

$$\frac{dS}{dt} = \frac{dS_{internal}}{dt} + \frac{dS_{exchange}}{dt}$$

Here, $\frac{dS_{exchange}}{dt}$ represents the exchange of information and energy with the environment. The very existence of this term is evidence that Spirit is not a closed "mind" but an open "field."

3. Structural Interpretation

While the physical framework allows us to define and measure the Spirit manifold, understanding its internal structure requires an interpretative model. We use the concepts of analytical psychology developed by C.G. Jung as a model to interpret the physical structures we measure.

- **Complex**: An individual's personal implementation of an Archetype, acting as an energy-charged node in the informational space.
- **Archetype**: A fundamental, structural template residing in the Collective Unconscious. This is an informational structure shared across humanity, beyond the "individual," supporting the "field-like" nature of Spirit.
- **Shadow**: The unconscious and often suppressed part of a Complex, a primary source of internal conflict.

Classification of Observed Patterns

We classify the observed patterns within the Spirit manifold into two primary categories:

- **Spirit Type (Integrated)**: Stable and integrated structures formed by the harmonious integration of Archetypes.

$$SpiritType = Archetype(Gene, Meme, Field)$$

- **Ghost Pattern (Unintegrated)**: Problematic structures arising from the interference of the Shadow and unintegrated Archetypes.

$$GhostPattern = f(Shadow, CollectiveArchetype, Meme, Field)$$

4. Measurement Methodology

We use a modified version of the Word Association Experiment (Jung, 1910) to probe the structure of an individual's information space. The probability of association is modeled as:

$$P(w_O|w_I) = \frac{\exp(\vec{w}_I \cdot \vec{w}_O) \cdot [r(w_I, w_O)]^\alpha \cdot \exp(\gamma \frac{\Delta SP}{\lambda}) \cdot \exp(\eta F)}{\sum_j \exp(\vec{w}_I \cdot \vec{w}_j) \cdot [r(w_I, w_j)]^\alpha \cdot \exp(\gamma \frac{\Delta SP}{\lambda}) \cdot \exp(\eta F)}$$

- **Semantic Component ($\vec{w}_I \cdot \vec{w}_O$):** Cosine similarity between word vectors.
- **Behavioural Component ($r(w_I, w_O)$):** Inverse of reaction time.
- **Physiological Component (Emotion F, Arousal ΔSP):** Multi-modal data from Hume AI and SKINPRO.

5. Methods

- **Participants:** Healthy adults (n=30) meeting specific inclusion/exclusion criteria.
- **Equipment:** High-resolution display, SKINPRO (8-channel skin potential), Hume AI Expression Measurement API (face, voice, language).
- **Data Integration:** Time-series synchronization of stimulus presentation, verbal responses, and physiological/emotional data with a $\pm$2-second matching window.
- **Analysis Pipeline:** 1024-dimensional complex space vectors reduced via PCA/UMAP for 3D visualization and classification using K-means and Isolation Forest.

6. Results

- **Classification Accuracy:** 82% (For Spirit Type identification)
- **Ghost Patterns Identified:** 14 (Distinct types identified)
- **Manifold Dim.:** 1024 (Initial feature space)

Spirit Type Distribution

Typical patterns composed of Gene + Meme + Field. - Hero Archetype: 124 responses - Sage Archetype: 98 responses - Lover Archetype: 76 responses - Caregiver Archetype: 45 responses Mean distance to archetype: 0.142

Ghost Pattern Detection

Hidden patterns primarily composed of Meme + Field. - Individual Shadow: 56 responses - Collective Unconscious Meme: 32 responses

Problematic Indicators: High Meme Variance, Pattern Interference, Cognitive Bias

The experimental results support the viability of our framework. We observed clear differentiation between stable archetypal structures (Spirit Types) and unstable shadow-driven interferences (Ghost Patterns).

7. Discussion

The successful construction of the high-dimensional manifold from linguistic, behavioural, and physiological data demonstrates the viability of our framework. The distinction between Spirit Types and Ghost Patterns provides a quantitative basis for identifying both healthy, coherent informational structures and problematic, bug-producing ones.

Our analysis suggests Ghost Patterns arise from measurable interference between different informational layers, such as conflicts between unconscious patterns (the Shadow) and conscious intentions. This work builds upon Jung's foundational work by embedding it within a modern information-theoretic and thermodynamic framework.

Comparison with Existing Research

- **Jung (1910):** We extend qualitative insights into quantitative structural analysis.
- **Landauer (1991):** Physical validation of information as a state variable for Spirit.
- **Botvinick (1998):** Rubber hand illusion as a manifold boundary transition.

Conclusion

This paper introduces a paradigm shift, moving “spirit” from metaphysical abstraction to concrete, physically measurable quantity.

As a central claim, we **negate the concept of the “individual” (the undividable)**. The human “mind” is not an internal entity separate from the body or environment, but an **open informational field that interpenetrates fields, physics, and others**. This cognitive shift is the reason we adopt the term “Spirit” rather than “Mind,” and it constitutes the ontological foundation of this research.

By operationalizing Spirit as a thermodynamic information quantity on a high-dimensional manifold, we have provided a framework that is both theoretically coherent and experimentally verifiable. This work lays the foundation for a new physics of spirit, opening the door to a truly empirical investigation of consciousness and the human condition.

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Additional Research: High-IQ Japanese GWAS

Leveraging Japan’s unique genetics, a GWAS targeting individuals with IQ 140 will compare genetic and cognitive data to identify SNPs linked to intelligence. The study begins in 2024 with results slated for publication. **Dataset:** 92 people / CAMS IQ140 sd15 - IQ180t / SNPs. **ref:** Jonathan R. I. Coleman et al, Mol Psychiatry 24, 182-197 (2019)